A Data Science Analysis

INDIAN FLIGHT PRICE PREDICTION MODELING

Building Predictive Models to Understand and Forecast Airfare Costs in India

- Exploratory Data Analysis
- Model Comparison
- Price Prediction

Project Proposal

✈ INDIAN FLIGHT PRICE PREDICTION

Data Science Capstone — Deliverable 1: Ideation & Proposal

Author	Bello-ososo Gabriel
Domain	Business & Finance Analytics
Dataset	Indian Airline Flight Price Dataset (Kaggle, 2019)
Task	Regression — Predict ticket price (INR) from flight attributes
Tools	Python · Pandas · Scikit-learn · Matplotlib · Seaborn

1. Problem Statement

The Indian domestic aviation market is one of the fastest-growing in the world, with over 140 million passengers annually. Ticket prices are notoriously volatile — the same seat can cost 2× to 5× more depending on when you book, which airline you choose, how many stops your route has, and what time of day you depart.

For travelers, this opacity creates frustration and inefficiency. For airlines and travel platforms, price prediction is a key operational tool enabling revenue optimization, dynamic pricing, and competitor benchmarking.

> Core Question: What structural factors drive Indian domestic flight prices, and how accurately
> can they be predicted from publicly available booking metadata alone?

1.1 Why It Matters

- Consumer value: A well-calibrated model helps travelers identify overpriced fares and choose optimal booking windows.
- Business value: Airlines and Online Travel Agencies (OTAs) use similar models for revenue management and dynamic pricing.
- Data richness: Real booking data contains irregular patterns (stop layovers, next-day arrivals, multi-carrier routes) that test robust data engineering skills.

2. Objectives & Research Questions

2.1 Project Objectives

1. Build a regression model predicting ticket price (INR) with $R^2 \geq 0.80$ and MAE ≤ ₹1,200.
2. Identify and rank the most influential structural drivers of price variation using feature importance analysis.
3. Produce actionable insights for travelers, OTAs, and airline operations teams.
4. Demonstrate production-quality data engineering on a real, messy dataset with documented cleaning decisions.

2.2 Research Questions

- Which features (airline, route, stops, departure time) drive the most price variation?
- Do non-stop flights always cost less than multi-stop flights?
- Does departure time of day (morning vs. evening vs. night) significantly affect ticket price?
- Which airline commands the most consistent price premium above the market median?
- Can a machine learning model predict flight prices within ₹1,000 MAE?

3. Proposed Data Sources

The dataset used in this project is a real-world scrape of Indian domestic flight booking data collected in 2019. It is publicly available on Kaggle under an open licence.

Attribute	Detail
Name	Flight Price Prediction Dataset
Source	Kaggle — Indian Domestic Flights (2019)
Raw Records	10,660 flight records
After Cleaning	8,764 records retained
Time Period	March – June 2019
Airlines Covered	8: IndiGo, Jet Airways, Air India, Multiple Carriers, SpiceJet, Vistara, GoAir, Air Asia
Source Cities	5: Delhi, Kolkata, Banglore, Mumbai, Chennai
Destination Cities	6: Cochin, Banglore, New Delhi, Kolkata, Delhi, Hyderabad
Raw Features	Airline, Date, Source, Destination, Route, Departure Time, Arrival Time, Duration, Total Stops, Additional Info, Price
Target Variable	Price (INR) — continuous numeric
License	Public / Kaggle Open Dataset

3.1 Data Quality Notes

The dataset is genuinely messy, which creates substantial data engineering opportunities:

- Duration stored as strings: '5h 25m', '19h', '25m' — requires custom parsing to integer minutes.
- Arrival times with next-day suffixes: '04:25 10 Jun' — time portion must be extracted robustly.
- Date format 'D/MM/YYYY' (e.g., '9/06/2019') — requires day first=True parsing.
- Total Stops as text categories ('non-stop', '1 stop', etc.) — mapped to integers 0–4.
- Additional info with 10+ messy categories — converted to 4 meaningful binary feature flags.
- Missing values across all columns — target rows dropped; others imputed with median or mode.

4. Evaluation Metrics

This is a regression task. Performance is evaluated using four complementary metrics:

Metric	Target	Justification
R^2 (Test Set)	≥ 0.80	Model must explain ≥ 80% of ticket price variance to be considered useful.
MAE (Test Set)	≤ ₹1,200	A ₹1,200 error on a ₹9,000 median ticket is a ~13% threshold — practically meaningful.
CV R^2 Std Dev	< 0.03	Low standard deviation across 5 folds ensures the model is stable and not overfit.
RMSE	Minimize	RMSE penalizes large prediction errors more than MAE — important for high-value tickets.

All models are evaluated with a fixed 80/20 stratified train-test split (random state = 42) and 5-fold cross-validation on the training set to ensure reliable, reproducible performance estimates.

5. Risks & Limitations

5.1 Data Risks

- Booking lead time absent: The number of days before departure is not in the dataset, yet it is the single most important variable in airline dynamic pricing. Its absence limits absolute accuracy.
- Single season only: The dataset covers March–June 2019. Monsoon season (July–September) and winter holidays are not represented. Seasonal generalization requires additional data.
- Class of service inferred: Economy vs. Business must be inferred from the Additional Info field rather than from a dedicated column, introducing potential label noise.
- Data vintage: Prices from 2019 may not reflect current fare structures. Jet Airways, the highest-priced carrier in the dataset, has since ceased operations.

5.2 Modelling Risks

- No outlier removal: Price outliers (e.g., ₹54,826) are retained. Their presence could be justified by legitimate late-booking premium or business-class factors. Removal would bias the model.
- LabelEncoder ordinality: Encoding airlines as integers imposes a false ordinal structure. Target encoding or one-hot encoding would be more appropriate for production.
- Sparse high-price segment: Very few flights exceed ₹40,000, limiting model accuracy in the luxury tier.

5.3 Assumptions

- Prices are single-leg fares (no return journey component).
- All 8,764 retained records represent genuine commercial transactions.
- Prices are quoted in Indian Rupees (INR) and represent economy class unless the business class flag is set.

Full analysis delivered in the accompanying Jupyter Notebook and Final Report.